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The state of the science for potential contributors to musculoskeletal injury following concussion: Mechanisms, gaps, and clinical considerations

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ABSTRACT

Concussion is a common mild traumatic brain injury affecting athletic, military, and general populations. While unrestricted medical clearance often occurs within 1-month, emerging evidence indicates prolonged sensorimotor control deficiencies and greater musculoskeletal injury risk after a concussion. Various theoretical frameworks and original studies indicate potential neurophysiological alterations affecting sensory input, sensorimotor integration, and neuromotor output, though the exact mechanisms remain unclear. This commentary aims to briefly review the literature on contributors to sensory and neuromotor deficiencies potentially related to musculoskeletal injury risk, recommend future research avenues to better understand these factors, and highlight current and future clinical applications.

1. Introduction

Concussion is a prominent injury in athletics, the military, and the general population (Chandran et al., 2022; Eastwood et al., 2023; Le et al., 2023; Zogg et al., 2018). This mild traumatic brain injury leads to transient symptom presentation, impaired neurocognitive and balance function, and often visual and/or vestibular dysfunction (Patricios et al., 2023). Though concussion medical clearance commonly occurs within approximately 28 days (Broglio et al., 2022), there is growing evidence of lingering sensorimotor control deficiencies (Howell et al., 2018a). Sensorimotor control is broadly defined in this context as the integration of external and internal stimuli to produce purposeful and coordinated movement, and is not typically assessed beyond static evaluations (i.e., postural stability) within the traditional concussion assessment battery (Lempke et al., 2020a, 2024a). Evidence of broad post-concussion sensorimotor control abnormalities includes deficient gait, especially when assessed under dual-task conditions (concurrent gait and cognitive task; Howell et al., 2018b; Buttner et al., 2020), altered biomechanics during functional tasks (Shumski et al., 2024; Avedesian et al., 2020; Lynall et al., 2018; Lapointe et al., 2018), and heightened lower extremity musculoskeletal injury risk for years even after the concussed patient is deemed clinically recovered (Lynall et al., 2015, 2017; McPherson et al., 2019; Jildeh et al., 2022; Kardouni et al., 2018). The estimated two-fold increased lower extremity musculoskeletal injury risk has received attention in the scientific literature over the last decade, with several systematic reviews and original research articles revealing this risk in athletic, general, and military populations (Lynall et al., 2015, 2017; McPherson et al., 2019; Jildeh et al., 2022; Kardouni et al., 2018).

It appears a relatively consistent relationship is present between concussion and a subsequent elevated lower extremity musculoskeletal injury risk (Lynall et al., 2015, 2017; McPherson et al., 2019; Jildeh et al., 2022; Kardouni et al., 2018), but the key question now is why? Why are individuals at increased lower extremity musculoskeletal injury risk when their concussion was deemed resolved? What factors are leading to this heightened risk, and how do we evaluate them? What rehabilitative strategies can we employ to potentially reduce this risk? The purpose of this commentary is to briefly review the literature on contributions to sensorimotor control deficiencies potentially related to increased lower extremity musculoskeletal injury risk and highlight current and future clinical applications while considering future avenues for research.

2. Potential contributors to post-concussion musculoskeletal injury risk

Multiple theoretical frameworks exist to describe potentially relevant factors and explanations for elevated post-concussion lower extremity musculoskeletal injury risk (Howell et al., 2018a; Eagle et al., 2020; Wilkerson et al., 2017). These frameworks broadly describe a neurophysiologic phenomenon that is altering 1) sensory information signaling (e.g., somatosensory, visual, auditory), 2) sensorimotor integration, cognitive processing, and subsequent action signaling, and/or 3) neuromotor output at the cortical, spinal, or peripheral level affecting

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movement capabilities. However, the specific neurophysiologic alteration(s) at play remain elusive with many areas warranting deeper investigation. We provide a conceptual figure highlighting example areas necessitating thorough examination in order to mechanistically decipher whether any neurophysiologic factors are attributed to greater lower extremity musculoskeletal injury risk after concussion (Fig. 1).

Original research has begun to isolate and evaluate sensory and motor domains not currently assessed in standard concussion evaluations, and attempt to relate these domains to lower extremity musculoskeletal injury risk. Sport-relevant biomechanical evaluations via kinematics and kinetics have broadly identified movement pattern differences such as subtly altered lower extremity joint range of motion (Shumski et al., 2024; Avedesian et al., 2020; Lapointe et al., 2018; Lee et al., 2021), trunk or center of mass displacement (Shumski et al., 2024; Lynall et al., 2018; Lapointe et al., 2018), and increased sagittal plane ankle and knee torque (Lee et al., 2021; Lempke et al., 2023a) for acutely concussed and concussion history cohorts relative to controls. However, the cumulative post-concussion biomechanics literature is heterogeneous regarding the movement protocols, outcomes, and time after concussion evaluated. This is an important consideration necessitating more consistent approaches to provide enhanced clarity (Fig. 1).

Numerous studies have evaluated corticospinal neuromotor function after concussion (Scott et al., 2020), but limited peripheral neuromuscular-focused post-concussion evaluations exist. Recent work has demonstrated no significant differences in quadricep or hamstring peak isokinetic torque (Chmielewski et al., 2023; Eagle et al., 2019) or quadricep peak maximal voluntary or involuntary (i.e., superimposed burst via electrical stimulation) isometric torque production (Lempke et al., 2023b). Muscle contractile properties have been observed to have lower magnitude and slower muscle contraction times (i.e., slower and lower contraction velocity) in the quadricep and hamstring muscle groups during controlled electrical stimulation (Kakavas et al., 2024). This may be a relevant biosignal underlying known reaction time deficits post-concussion (Lempke et al., 2020b, 2023a) and potentially relating to increased lower extremity musculoskeletal injury risk, but more work is needed with larger sample sizes to better understand what specific peripheral neuromotor mechanisms may be altered following concussion (Fig. 1).

A plethora of research evaluating broad post-concussion sensorimotor system function (i.e., somatosensory, vestibular, visual) via postural stability assessments exists (Murray et al., 2019). However, this work fundamentally differs from isolated somatosensory subsystem (i.e., pain, temperature, tactile, proprioception: [kinesthesia, joint position sense, sense of force]) evaluations as subtle subsystem changes could be masked through sensory reweighting if only measured through gross postural stability or gait evaluations (Asslander and Peterka, 2014). Unfortunately, investigation of isolated post-concussion somatosensory function in the lower extremity is limited to one study to our knowledge. The authors reported significantly less plantar touch sensation, non-significant but moderate effect size for less plantar pressure pain sensation, and no significant or notable effect sizes for knee passive joint repositioning acutely for concussed individuals relative to controls in this preliminary study (Lempke et al., 2023c). Given the paucity of somatosensory post-concussion research and limitations of the existing work, considerable work is needed to understand if and for how long these senses are altered following concussion (Fig. 1).

3. Clinical integration

Though the literature is clear on heightened lower extremity musculoskeletal injury risk after clinical concussion recovery, several critical gaps remain. Importantly, a direct causal relationship between concussion and lower extremity musculoskeletal risk has not been established. Evaluating for a causal relationship would require large scale, multi-site studies that includes longitudinal tracking of concussion and musculoskeletal outcomes over several years along with enrolling individuals with and without prior concussion and lower extremity musculoskeletal injury history. Though challenging, research such as this is key to confirming the link between concussion and musculoskeletal injury. These data would also help the field understand important patient characteristic factors, such as sport, sex, and injury type, that may play a critical role in mediating or moderating the relationship. Another key research gap is determining the specific sensorimotor control deficiencies that are most affected by concussion (Fig. 1). Again,

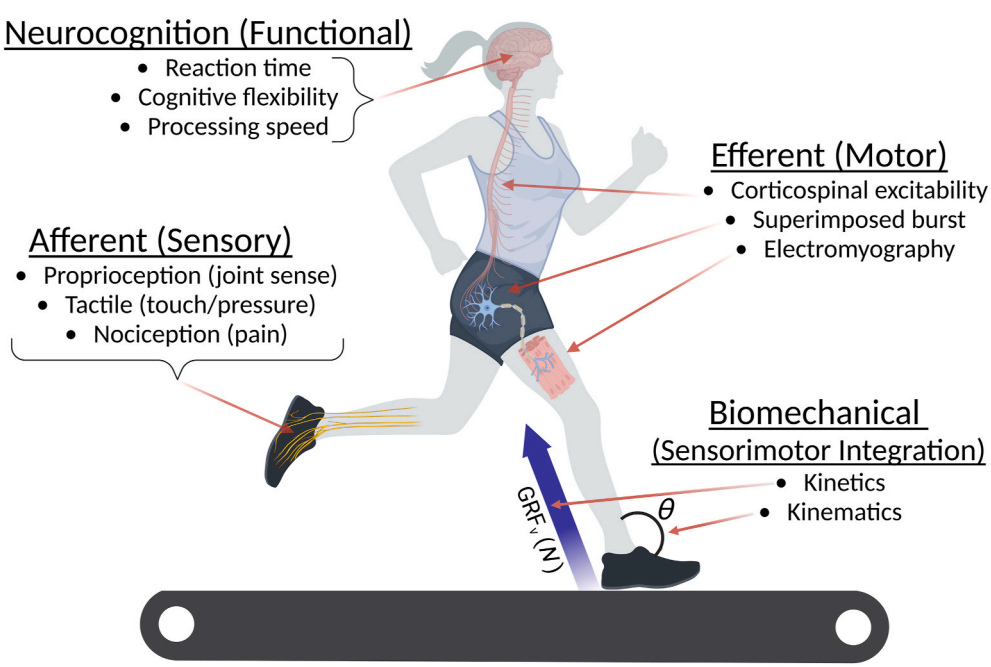

Fig. 1. Conceptual Model for Potential Targets to Address Lower Extremity Musculoskeletal Injury Following Concussion.
Fig. 1 legend. Theorized example domains and outcomes warranting targeted examination in high-quality research study designs in order to mechanistically understand contributors to lower extremity musculoskeletal injury following concussion.

multi-site, longitudinal studies are critical here, along with accurate and reliable measurement of important sensorimotor control outcomes, such as those described previously. Determining a causal relationship, if one does exist, and understanding the role of specific sensorimotor control outcomes are imperative to allow researchers and clinicians to design and implement effective interventions for post-concussion lower extremity musculoskeletal injury risk. Building such interventions into post-concussion management, both before and after the patient is cleared to return to normal activity, could be a significant step forward in concussion care.

Cumulatively, two pieces of the post-concussion sensorimotor control puzzle appear clear: 1) current concussion evaluations do not directly or indirectly evaluate musculoskeletal injury-relevant outcomes (Buckley et al., 2020), and 2) there remains heavy uncertainty surrounding which specific neurophysiologic functions are impaired short- and long-term after concussion that might attribute to heightened lower extremity musculoskeletal injury risk (Fig. 1). Though isolated evaluations are necessary to begin understanding mechanistic alterations after concussion, researchers have begun to bridge the clinical-research gap so clinicians may begin to intervene and improve patient outcomes. For example, researchers implemented an established musculoskeletal injury risk reduction program after concussion and observed 3.56 times lower musculoskeletal injury hazard among those who completed the program relative to the standard of care (Howell et al., 2022). The program focused on plyometric, strength, technique, and balance training and had been previously used outside of concussion in youth athletes to improve neuromuscular function and reduce sports injury (Foss et al., 2018; Myer et al., 2008). Additional work has started addressing the limited return to sport or activity applicability of concussion assessments by developing and employing new evaluation batteries focused on sport-related cognitive-motor and cardiovascular performance to serve as return to activity or medical clearance evaluations (Kochick et al., 2022; Lempke et al., 2023d, 2024b). These evolving batteries include sport-specific movements like jumping and cutting while assessing various outcomes such as reaction time.

In the meantime, clinicians should be aware of the possibility of post-concussion sensorimotor control deficiencies that are insufficiently assessed with their standard post-concussion assessment battery. Incorporating functional rehabilitation interventions that include dual-tasking and multi-sensory challenges may be beneficial without specific post-concussion protocols. For example, a common proprioceptive rehabilitation exercise includes throwing and catching a ball while balancing on an unstable surface. The clinician could add layers to this exercise by introducing cognitive challenges that increase in difficulty (spell 3-letter words backwards → alphabet backwards). Additionally, clinicians could integrate the motor and cognitive task by using a ball with different letters on it, and when the patient catches the ball, they immediately look at the first letter they see on the ball and complete a cognitive task based on that letter (A = count backwards by 7's from 100, B = months of the year backwards, etc.). If rehabilitation is conducted in a controlled environment by a trained clinician, negative consequences are highly unlikely. For the readers who function as clinicians, we support the modified adoption of established interventions into rehabilitation, such as dual-task (Register-et al., 2019), cognitive-behavioral (Porter et al., 2022), multi-sensory integration such as described above, and other rehabilitation techniques for post-concussion implementation. Given the plethora of data suggesting supervised post-concussion physical activity is safe (Gallow et al., 2024; Register-et al., 2024; Leddy et al., 2018), clinicians should not hesitate to incorporate symptom-limited post-concussion rehabilitation methodology in controlled, safe environments while researchers continue to explore the most effective ways to mitigate post-concussion lower extremity musculoskeletal injury risk.

CRediT authorship contribution statement

Landon B. Lempke: Writing – review & editing, Writing – original draft, Conceptualization. **Robert C. Lynall:** Writing – review & editing, Writing – original draft, Conceptualization.

4. Disclosures

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References

Asslander, L., Peterka, R.J., 2014. Sensory reweighting dynamics in human postural control. J. Neurophysiol. 111 (9), 1852–1864.

Avedesian, J.M., Covassin, T., Dufek, J.S., 2020. Landing biomechanics in adolescent athletes with and without a history of sports-related concussion. J. Appl. Biomech. 36 (5), 313–318.

Broglio, S.P., McAllister, T., Katz, B.P., et al., 2022. The natural history of sport-related concussion in collegiate athletes: findings from the NCAA-DoD CARE consortium. Sports Med. 52 (2), 403–415.

Buckley, T.A., Howard, C.M., Oldham, J.R., Lynall, R.C., Swanik, C.B., Getchell, N., 2020. No clinical predictors of postconcussion musculoskeletal injury in College athletes. Med. Sci. Sports Exerc. 52 (6), 1256–1262.

Buttner, F., Howell, D.R., Ardern, C.L., et al., 2020. Concussed athletes walk slower than non-concussed athletes during cognitive-motor dual-task assessments but not during single-task assessments 2 months after sports concussion: a systematic review and meta-analysis using individual participant data. Br. J. Sports Med. 54 (2), 94–101.

Chandran, A., Boltz, A.J., Morris, S.N., et al., 2022. Epidemiology of concussions in national collegiate athletic association (NCAA) sports: 2014/15-2018/19. Am. J. Sports Med. 50 (2), 526–536.

Chmielewski, T.L., Suzuki, S., Dubose, D., Herman, D.C., Clugston, J.R., Horodyski, M., 2023. Mild quadriceps and hamstring strength deficits do not persist after sport-related concussion. Sport Health 15 (4), 512–518.

Eagle, S.R., Kontos, A.P., Mi, Q.I., et al., 2019. Shared neuromuscular performance traits in military personnel with prior concussion. Med. Sci. Sports Exerc. 51 (8), 1619–1625.

Eagle, S.R., Kontos, A.P., Pepping, G.J., et al., 2020. Increased risk of musculoskeletal injury following sport-related concussion: a perception-action coupling approach. Sports Med. 50 (1), 15–23.

Eastwood, D., Owen, C., Phillips, G., et al., 2023. Incidence of concussion in men's Super League, Championship, and Academy rugby league matches between 2016 and 2022. J. Sci. Med. Sport/Sports Medicine Australia 26 (10), 539–544.

Foss, K.D.B., Thomas, S., Khoury, J.C., Myer, G.D., Hewett, T.E., 2018. A school-based neuromuscular training program and sport-related injury incidence: a prospective randomized controlled clinical trial. J. Athl. Train. 53 (1), 20–28.

Gallow, S., Beard, J., McGinley, J., Olver, J., Williams, G., 2024. Cardiorespiratory fitness assessment and training in the early sub-acute phase of recovery following traumatic brain injury: a systematic review. Brain Inj. 1–12.

Howell, D.R., Lynall, R.C., Buckley, T.A., Herman, D.C., 2018a. Neuromuscular control deficits and the risk of subsequent injury after a concussion: a scoping review. Sports Med. 48 (5), 1097–1115.

Howell, D.R., Buckley, T.A., Lynall, R.C., Meehan 3rd, W.P., 2018b. Worsening dual-task gait costs after concussion and their association with subsequent sport-related injury. J. Neurotrauma 35 (14), 1630–1636.

Howell, D.R., Seehusen, C.N., Carry, P.M., Walker, G.A., Reinking, S.E., Wilson, J.C., 2022. An 8-week neuromuscular training program after concussion reduces 1-year subsequent injury risk: a randomized clinical trial. Am. J. Sports Med. 50 (4), 1120–1129.

Jildeh, T.R., Castle, J.P., Buckley, P.J., Abbas, M.J., Hegde, Y., Okoroha, K.R., 2022. Lower extremity injury after return to sports from concussion: a systematic review. Orthop J. Sports Med. 10 (1), 23259671211068438.

Kakavas, G., Tsiokanos, A., Potoupnis, M., Tsaklis, P.V., 2024. Mechanical and contractile properties of knee joint muscles after sports-related concussions in women footballers. J. Funct. Morphol. Kinesiol. 9 (2).

Kardouni, J.R., Shing, T.L., McKinnon, C.J., Scofield, D.E., Proctor, S.P., 2018. Risk for lower extremity injury after concussion: a matched cohort study in soldiers. J. Orthop. Sports Phys. Ther. 48 (7), 533–540.

Kochick, V., Sinnott, A.M., Eagle, S.R., et al., 2022. The dynamic exertion test for sport-related concussion: a comparison of athletes at return-to-play and healthy controls. Int. J. Sports Physiol. Perform. 17 (6), 834–843.
Lapointe, A.P., Nolasco, L.A., Sosnowski, A., et al., 2018. Kinematic differences during a jump cut maneuver between individuals with and without a concussion history. Int. J. Psychophysiol. 132 (Pt A), 93–98.
Le, T.D., Gurney, J.M., Singh, K.P., et al., 2023. Trends in traumatic brain injury among U.S. Service members deployed in Iraq and Afghanistan, 2002-2016. Am. J. Prev. Med. 65 (2), 230–238.
Leddy, J.J., Hinds, A.L., Miecznikowski, J., et al., 2018. Safety and prognostic utility of provocative exercise testing in acutely concussed adolescents: a randomized trial. Clin. J. Sport Med. 28 (1), 13–20.
Lee, B.J., Blueitt, D., Hannon, J., Goto, S., Garrison, C., 2021. Movement patterns during a jump-landing task in athletes after sport-related concussion and healthy control individuals. J. Athl. Train. 56 (12), 1306–1312.
Lempke, L.B., Schmidt, J.D., Lynall, R.C., 2020a. Athletic Trainers' concussion-assessment and concussion-management practices: an update. J. Athl. Train. 55 (1), 17–26.
Lempke, L.B., Howell, D.R., Eckner, J.T., Lynall, R.C., 2020b. Examination of reaction time deficits following concussion: a systematic review and meta-analysis. Sports Med. 50 (7), 1341–1359.
Lempke, L.B., Hoch, M.C., Call, J.A., Schmidt, J.D., Lynall, R.C., 2023a. Single-leg hop stabilization throughout concussion recovery: a preliminary biomechanical assessment. J. Sport Rehabil. 32 (5), 513–523.
Lempke, L.B., Call, J.A., Hoch, M.C., Schmidt, J.D., Lynall, R.C., 2023b. Lower-extremity neuromuscular function following concussion: a preliminary examination. J. Sport Rehabil. 32 (1), 31–39.
Lempke, L.B., Hoch, M.C., Call, J.A., Schmidt, J.D., Lynall, R.C., 2023c. Lower extremity somatosensory function throughout concussion recovery: a prospective cohort study. J. Head Trauma Rehabil. 38 (2), E156–E166.
Lempke, L.B., Shumski, E.J., Prato, T.A., Lynall, R.C., 2023d. Reliability and minimal detectable change of the standardized assessment of reaction time. J. Athl. Train. 58 (6), 579–587.
Lempke, L.B., Boltz, A.J., Garcia, G.P., et al., 2024a. Optimizing baseline and post-concussion assessments through identification, confirmation, and equivalence of latent factor structures: findings from the NCAA-DoD CARE Consortium. Clin. Neuropsychol. 38 (5), 1156–1174.
Lempke, L.B., Terry, D.P., Meehan, W.P., Brett, B.L., 2024b. Test-retest reliability and minimal detectable change of an ecologically valid return to activity evaluation: the functional assessment of neurocognition in sport (FANS). J. Athl. Train. 59 (6s).
Lynall, R.C., Mauntel, T.C., Padua, D.A., Mihalik, J.P., 2015. Acute lower extremity injury rates increase after concussion in College athletes. Med. Sci. Sports Exerc. 47 (12), 2487–2492.
Lynall, R.C., Mauntel, T.C., Pohlig, R.T., et al., 2017. Lower extremity musculoskeletal injury risk after concussion recovery in high school athletes. J. Athl. Train. 52 (11), 1028–1034.
Lynall, R.C., Blackburn, J.T., Guskiewicz, K.M., Marshall, S.W., Plummer, P., Mihalik, J. P., 2018. Reaction time and joint kinematics during functional movement in recently concussed individuals. Arch. Phys. Med. Rehabil. 99 (5), 880–886.
McPherson, A.L., Nagai, T., Webster, K.E., Hewett, T.E., 2019. Musculoskeletal injury risk after sport-related concussion: a systematic review and meta-analysis. Am. J. Sports Med. 47 (7), 1754–1762.
Murray, N.G., Reed-Jones, R.J., Szekely, B.J., Powell, D.W., 2019. Clinical assessments of balance in adults with concussion: an update. Semin. Speech Lang. 40 (1), 48–56.
Myer, G.D., Chu, D.A., Brent, J.L., Hewett, T.E., 2008. Trunk and hip control neuromuscular training for the prevention of knee joint injury. Clin. Sports Med. 27 (3), 425–448 ix.
Patricios, J.S., Schneider, K.J., Dvorak, J., et al., 2023. Consensus statement on concussion in sport: the 6th international conference on concussion in sport-amsterdam, october 2022. Br. J. Sports Med. 57 (11), 695–711.
Porter, K.H., Quintana, C., Morelli, N., et al., 2022. Neurocognitive function influences dynamic postural stability strategies in healthy collegiate athletes. J. Sci. Med. Sport/Sports Medicine Australia 25 (1), 64–69.
Register-Mihalik, J.K., Guskiewicz, K.M., Marshall, S.W., et al., 2019. Methodology and implementation of a randomized controlled trial (RCT) for early post-concussion rehabilitation: the active rehab study. Front. Neurol. 10, 1176.
Register-Mihalik, J.K., Guskiewicz, K.M., Marshall, S.W., et al., 2024. Symptom exacerbation and adverse events during a randomized trial of early-stage concussion rehabilitation. J. Athl. Train. https://meridian.allenpress.com/jat/article/doi/10.4085/1062-6050-0696.23/500881/Symptom-Exacerbation-and-Adverse-Events-during-a.
Scott, E., Kidgell, D.J., Frazer, A.K., Pearce, A.J., 2020. The neurophysiological responses of concussive impacts: a systematic review and meta-analysis of transcranial magnetic stimulation studies. Front. Hum. Neurosci. 14, 306.
Shumski, E.J., Oh, J., Schmidt, J.D., Lynall, R.C., 2024. Trunk and lower extremity biomechanics in female athletes with and without a concussion history. J. Athl. Train. 59 (7), 751–761.
Wilkerson, G.B., Grooms, D.R., Acocello, S.N., 2017. Neuromechanical considerations for postconcussion musculoskeletal injury risk management. Curr. Sports Med. Rep. 16 (6), 419–427.
Zogg, C.K., Haring, R.S., Xu, L., et al., 2018. Patient presentations in outpatient settings: epidemiology of adult head trauma treated outside of hospital emergency departments. Epidemiology 29 (6), 885–894.